1) a)

$$|\mathbf{A}| = \begin{vmatrix} 4 & 2 & 6 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{vmatrix}$$

$$= 4 \times (6 \times 12 - 15 \times 0) - 2 \times (0 \times 12 - 15 \times 0) + 6 \times (0 \times 0 - 6 \times 0)$$

$$= 4 \times 6 \times 12$$

$$= 288$$

1) b)

$$2\begin{vmatrix} 2 & 1 & 3 \\ 0 & 6 & 15 \\ 0 & 0 & 12 \end{vmatrix}$$

$$= 2\big(2 \times (6 \times 12 - 15 \times 0) - 1 \times (0 \times 12 - 15 \times 0) + 3 \times (0 \times 0 - 6 \times 0)\big)$$

$$= 2(2 \times 6 \times 12)$$

$$= 288$$

1) c)

$$6\begin{vmatrix} 2 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix}$$

$$= 6\big(2 \times (6 \times 4 - 5 \times 0) - 1 \times (0 \times 4 - 5 \times 0) + 1 \times (0 \times 0 - 6 \times 0)\big)$$

$$= 6(2 \times 6 \times 4)$$

$$= 288$$

1) d)

$$12\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 5 \\ 0 & 0 & 4 \end{vmatrix}$$

$$= 12\big(1 \times (6 \times 4 - 5 \times 0) - 1 \times (0 \times 4 - 5 \times 0) + 1 \times (0 \times 0 - 6 \times 0)\big)$$

$$= 12(1 \times 6 \times 4)$$

$$= 288$$

1) e)

$$48\begin{vmatrix}1 & 1 & 1\\0 & 6 & 5\\0 & 0 & 1\end{vmatrix}$$

$$= 48\big(1 \times (6 \times 1 - 5 \times 0) - 1 \times (0 \times 1 - 5 \times 0) + 1 \times (0 \times 0 - 6 \times 0)\big)$$

$$= 48(1 \times 6 \times 1)$$

$$= 288$$

2) a)

$$|\mathbf{A}| = \begin{vmatrix}22 & 0 & 0\\660 & 15 & 0\\220 & -12 & 2\end{vmatrix}$$

$$= 22 \times \big(15 \times 2 - 0 \times (-12)\big) - 0 \times (660 \times 2 - 0 \times 220) + 0 \times (660 \times (-12) - 15 \times 220)$$

$$= 22 \times 15 \times 2$$

$$= 660$$

2) b)

$$2\begin{vmatrix}22 & 0 & 0\\660 & 15 & 0\\220 & -12 & 1\end{vmatrix}$$

$$= 2\Big(22 \times \big(15 \times 1 - 0 \times (-12)\big) - 0 \times (660 \times 1 - 0 \times 220) + 0 \times (660 \times (-12) - 15 \times 220)\Big)$$

$$= 2(22 \times 15 \times 1)$$

$$= 660$$

2) c)

$$4\begin{vmatrix}11 & 0 & 0\\330 & 15 & 0\\110 & -12 & 1\end{vmatrix}$$

$$= 4\Big(11 \times \big(15 \times 1 - 0 \times (-12)\big) - 0 \times (330 \times 1 - 0 \times 110) + 0 \times (330 \times (-12) - 15 \times 110)\Big)$$

$$= 4(11 \times 15 \times 1)$$

$$= 660$$

2) d)

$$12\begin{vmatrix} 11 & 0 & 0 \\ 110 & 5 & 0 \\ 110 & -12 & 1 \end{vmatrix}$$

$$= 12\Big(11 \times \big(5 \times 1 - 0 \times (-12)\big) - 0 \times (110 \times 1 - 0 \times 110) + 0 \times (110 \times (-12) - 5 \times 110)\Big)$$

$$= 12(11 \times 5 \times 1)$$

$$= 660$$

2) e)

$$60\begin{vmatrix} 11 & 0 & 0 \\ 22 & 1 & 0 \\ 110 & -12 & 1 \end{vmatrix}$$

$$= 60\Big(11 \times \big(1 \times 1 - 0 \times (-12)\big) - 0 \times (22 \times 1 - 0 \times 110) + 0 \times (22 \times (-12) - 1 \times 110)\Big)$$

$$= 60(11 \times 1 \times 1)$$

$$= 660$$

2) f)

$$660\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 10 & -12 & 1 \end{vmatrix}$$

$$= 660\Big(1 \times \big(1 \times 1 - 0 \times (-12)\big) - 0 \times (2 \times 1 - 0 \times 10) + 0 \times (2 \times (-12) - 1 \times 10)\Big)$$

$$= 660(1 \times 1 \times 1)$$

$$= 660$$

2) g)

$$|\mathbf{A}| = \begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 2 \end{vmatrix}$$

$$= 2\begin{vmatrix} 22 & 0 & 0 \\ 660 & 15 & 0 \\ 220 & -12 & 1 \end{vmatrix}$$

$$= 2 \times 2\begin{vmatrix} 11 & 0 & 0 \\ 330 & 15 & 0 \\ 110 & -12 & 1 \end{vmatrix}$$

$$= 2 \times 2 \times 3\begin{vmatrix} 11 & 0 & 0 \\ 110 & 5 & 0 \\ 110 & -12 & 1 \end{vmatrix}$$

$$= 2 \times 2 \times 3 \times 5\begin{vmatrix} 11 & 0 & 0 \\ 22 & 1 & 0 \\ 110 & -12 & 1 \end{vmatrix}$$

$$= 2 \times 2 \times 3 \times 5 \times 11\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 10 & -12 & 1 \end{vmatrix}$$

$$= 2 \times 2 \times 3 \times 5 \times 11 \times 1$$

$$\therefore |\mathbf{A}| = 660 = 2^2 \times 3 \times 5 \times 11$$

3) a)

$$|\mathbf{A}| = \begin{vmatrix} k & 3k^3 & 4k^3(k-1) \\ 0 & k^2 & 2k^2(k-1) \\ 0 & 0 & 3k^2(k-1) \end{vmatrix}$$

$$= k \times (k^2 \times 3k^2(k-1) - 2k^2(k-1) \times 0) - 3k^3 \times (0 \times 3k^2(k-1) - 2k^2(k-1) \times 0) + 4k^3(k-1) \times (0 \times 0 - k^2 \times 0)$$

$$= k \times k^2 \times 3k^2(k-1)$$

$$= 3k^6 - 3k^5$$

$$= 3k^5(k-1)$$

3) b)

$$k\begin{vmatrix}1 & 3k^2 & 4k^2(k-1)\\ 0 & k^2 & 2k^2(k-1)\\ 0 & 0 & 3k^2(k-1)\end{vmatrix}$$

$$= k\big(1\times(k^2\times 3k^2(k-1)-2k^2(k-1)\times 0)-3k^2\times(0\times 3k^2(k-1)-2k^2(k-1)\times 0)+4k^2(k-1)\times(0\times 0-k^2\times 0)\big)$$

$$= k\big(1\times k^2\times 3k^2(k-1)\big)$$

$$= 3k^6-3k^5$$

$$= 3k^5(k-1)$$

3) c)

$$k(k-1)\begin{vmatrix}1 & 3k^2 & 4k^2\\ 0 & k^2 & 2k^2\\ 0 & 0 & 3k^2\end{vmatrix}$$

$$= k(k-1)\big(1\times(k^2\times 3k^2-2k^2\times 0)-3k^2\times(0\times 3k^2-2k^2\times 0)+4k^2\times(0\times 0-k^2\times 0)\big)$$

$$= k(k-1)(1\times k^2\times 3k^2)$$

$$= 3k^6-3k^5$$

$$= 3k^5(k-1)$$

3) d)

$$k^3(k-1)\begin{vmatrix}1 & 3k^2 & 4\\ 0 & k^2 & 2\\ 0 & 0 & 3\end{vmatrix}$$

$$= k^3(k-1)\big(1\times(k^2\times 3-2\times 0)-3k^2\times(0\times 3-2\times 0)+4\times(0\times 0-k^2\times 0)\big)$$

$$= k^3(k-1)(1\times k^2\times 3)$$

$$= 3k^6-3k^5$$

$$= 3k^5(k-1)$$

3) e)

$$k^5(k-1)\begin{vmatrix}1 & 3 & 4\\ 0 & 1 & 2\\ 0 & 0 & 3\end{vmatrix}$$

$$= k^5(k-1)\big(1\times(1\times 3-2\times 0)-3\times(0\times 3-2\times 0)+4\times(0\times 0-1\times 0)\big)$$

$$= k^5(k-1)(1\times 1\times 3)$$

$$= 3k^6-3k^5$$

$$= 3k^5(k-1)$$

3) f)

$|\mathbf{A}| = 3k^5(k-1)$

4) a)

$$|\mathbf{A}| = \begin{vmatrix} 5k-10 & 0 & 0 \\ 4k-8 & 6-2k & 0 \\ 12-6k & 5k-15 & 3k-18 \end{vmatrix}$$

$= (5k-10)\big((6-2k)(3k-18) - 0 \times (5k-15)\big) - 0 \times \big((4k-8)(3k-18) - 0 \times (12-6k)\big) + 0 \times \big((4k-8)(5k-15) - (6-2k)(12-6k)\big)$

$= (5k-10)(6-2k)(3k-18)$

4) b)

$$|\mathbf{A}| = (k-2)\begin{vmatrix} 5 & 0 & 0 \\ 4 & 6-2k & 0 \\ -6 & 5k-15 & 3k-18 \end{vmatrix}$$

$= (k-2)\Big(5\big((6-2k)(3k-18) - 0 \times (5k-15)\big) - 0 \times \big(4(3k-18) - 0 \times (-6)\big) + 0 \times \big(4(5k-15) - (6-2k)(-6)\big)\Big)$

$= (k-2) \times 5(6-2k)(3k-18)$

$= 5(k-2)(6-2k)(3k-18)$

4) c)

$$|\mathbf{A}| = (k-2)(k-3)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3k-18 \end{vmatrix}$$

$= (k-2)(k-3)\Big(5\big((-2)(3k-18) - 0 \times 5\big) - 0 \times \big(4(3k-18) - 0 \times (-6)\big) + 0 \times \big(4(5) - (-2)(-6)\big)\Big)$

$= (k-2)(k-3) \times 5(-2)(3k-18)$

$= -10(k-2)(k-3)(3k-18)$

4) d)

$$|\mathbf{A}| = (k-2)(k-3)(k-6)\begin{vmatrix} 5 & 0 & 0 \\ 4 & -2 & 0 \\ -6 & 5 & 3 \end{vmatrix}$$

$$= (k-2)(k-3)(k-6)\Big(5\big((-2)(3) - 0\times 5\big) - 0\times\big(4(3) - 0\times(-6)\big) + 0\times\big(4(5) - (-2)(-6)\big)\Big)$$

$$= (k-2)(k-3)(k-6)\times 5(-2)(3)$$

$$= -30(k-2)(k-3)(k-6)$$

4) e)

$$|\mathbf{A}| = -30(k-2)(k-3)(k-6)$$

5)

$$|\mathbf{A}| = \begin{vmatrix} 24k & 4k+4 & 3k-6 \\ -6k & 2k+2 & 2k-4 \\ 12k & -3k-3 & 8-4k \end{vmatrix}$$

Factor $6k$ out of C1:

$$= 6k\begin{vmatrix} 4 & 4(k+1) & 3(k-2) \\ -1 & 2(k+1) & 2(k-2) \\ 2 & -3(k+1) & -4(k-2) \end{vmatrix}$$

Factor $(k+1)$ out of C2:

$$= 6k(k+1)\begin{vmatrix} 4 & 4 & 3(k-2) \\ -1 & 2 & 2(k-2) \\ 2 & -3 & -4(k-2) \end{vmatrix}$$

Factor $(k-2)$ out of C3:

$$= 6k(k+1)(k-2)\begin{vmatrix} 4 & 4 & 3 \\ -1 & 2 & 2 \\ 2 & -3 & -4 \end{vmatrix}$$

Then:

$$= 6k(k+1)(k+2)\Big(4\big(2\times(-4) - 2\times(-3)\big) - 4\big((-1)\times(-4) - 2\times 2\big) + 3\big((-1)\times(-3) - 2\times 2\big)\Big)$$

$$= 6k(k+1)(k+2)\big(4(-2) - 4(0) + 3(-1)\big)$$

$$= 6k(k+1)(k+2)(-8 - 0 - 3)$$

$$= -66k(k+1)(k+2)$$

6)

$$|\mathbf{A}| = 0 \Rightarrow \begin{vmatrix} 4k-2 & 8k-4 & 5-10k \\ 3-9k & 18k-6 & 6k-2 \\ -1-4k & 12k+3 & 20k+5 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 2(2k-1) & 4(2k-1) & -5(2k-1) \\ -3(3k-1) & 6(3k-1) & 2(3k-1) \\ -(4k+1) & 3(4k+1) & 5(4k+1) \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)\begin{vmatrix} 2 & 4 & -5 \\ -3(3k-1) & 6(3k-1) & 2(3k-1) \\ -(4k+1) & 3(4k+1) & 5(4k+1) \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)(3k-1)\begin{vmatrix} 2 & 4 & -5 \\ -3 & 6 & 2 \\ -(4k+1) & 3(4k+1) & 5(4k+1) \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)(3k-1)(4k+1)\begin{vmatrix} 2 & 4 & -5 \\ -3 & 6 & 2 \\ -1 & 3 & 5 \end{vmatrix} = 0$$

$$\Rightarrow (2k-1)(3k-1)(4k+1)\Big(2(6\times 5-2\times 3)-4\big((-3)\times 5-2\times(-1)\big)-5\big((-3)\times 3-6\times(-1)\big)\Big) = 0$$

$$\Rightarrow (2k-1)(3k-1)(4k+1)\big(2(24)-4(-13)-5(-3)\big) = 0$$

$$\Rightarrow 115(2k-1)(3k-1)(4k+1) = 0$$

$$\Rightarrow k = \tfrac{1}{2} \text{ or } k = \tfrac{1}{3} \text{ or } k = -\tfrac{1}{4}$$

7) a)

$$|\mathbf{A}| = \begin{vmatrix} 0 & -1 & 3 \\ 2 & 2 & 3 \\ 2 & 1 & -3 \end{vmatrix}$$

$$= 0\times(2\times(-3)-3\times 1)-(-1)\times(2\times(-3)-3\times 2)+3\times(2\times 1-2\times 2)$$

$$= 0-(-1)(-12)+3(-2)$$

$$= -18$$

7) b)

$$|\mathbf{A}| = \begin{vmatrix} 0 & -1 & 3 \\ 2 & 2 & 3 \\ 2 & 1 & -3 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + \mathrm{R3}$:

$$|\mathbf{A}| = \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 3 \\ 2 & 1 & -3 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + \mathrm{R3}$:

$$|\mathbf{A}| = \begin{vmatrix} 2 & 0 & 0 \\ 4 & 3 & 0 \\ 2 & 1 & -3 \end{vmatrix}$$

7) c)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 0 & 0 \\ 4 & 3 & 0 \\ 2 & 1 & -3 \end{vmatrix} = 2 \times 3 \times (-3) = -18$$

8) a)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 2 & -1 \\ 2 & 5 & 0 \\ 2 & 5 & 4 \end{vmatrix}$$

$$= 2(5 \times 4 - 0 \times 5) - 2(2 \times 4 - 0 \times 2) - (2 \times 5 - 5 \times 2)$$

$$= 2(20) - 2(8) - 0$$

$$= 24$$

8) b)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 2 & -1 \\ 2 & 5 & 0 \\ 2 & 5 & 4 \end{vmatrix}$$

Factor 2 out of C1:

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 1 & 5 & 0 \\ 1 & 5 & 4 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R2}$:

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 1 & 5 & 0 \\ 0 & 0 & 4 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{vmatrix}$$

8) c)

$$|\mathbf{A}| = 2\begin{vmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{vmatrix} = 2 \times (1 \times 3 \times 4) = 2 \times 12 = 24$$

9) a)

$$\begin{vmatrix} 1 & 0 & 5 \\ 2 & 1 & 2 \\ 3 & 2 & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R1}$:

$$= \begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & -8 \\ 3 & 2 & 1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 3\mathrm{R1}$:

$$= \begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & -8 \\ 0 & 2 & -14 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 2\mathrm{R2}$:

$$= \begin{vmatrix} 1 & 0 & 5 \\ 0 & 1 & -8 \\ 0 & 0 & 2 \end{vmatrix}$$

Then:

$$= 1 \times 1 \times 2$$

$$= 2$$

9) b)

$$\begin{vmatrix} 2 & 4 & 6 \\ 3 & -1 & 1 \\ -2 & 2 & 1 \end{vmatrix}$$

Factor 2 out of R1:

$$= 2\begin{vmatrix} 1 & 2 & 3 \\ 3 & -1 & 1 \\ -2 & 2 & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 3\mathrm{R1}$:

$$= 2\begin{vmatrix} 1 & 2 & 3 \\ 0 & -7 & -8 \\ -2 & 2 & 1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + 2\mathrm{R1}$:

$$= 2\begin{vmatrix} 1 & 2 & 3 \\ 0 & -7 & -8 \\ 0 & 6 & 7 \end{vmatrix}$$

Factor 7 out of R2:

$$= 14\begin{vmatrix} 1 & 2 & 3 \\ 0 & -1 & -\frac{8}{7} \\ 0 & 6 & 7 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + 6\mathrm{R2}$:

$$= 14\begin{vmatrix} 1 & 2 & 3 \\ 0 & -1 & -\frac{8}{7} \\ 0 & 0 & \frac{1}{7} \end{vmatrix}$$

Then:

$$= 14\left(1 \times (-1) \times \frac{1}{7}\right)$$

$$= -2$$

9) c)

$$\begin{vmatrix} 4 & 1 & 6 \\ -1 & 3 & 3 \\ 4 & -2 & 3 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 4\mathrm{R1}$:

$$= \begin{vmatrix} 4 & 1 & 6 \\ -1 & 3 & 3 \\ 0 & -3 & -3 \end{vmatrix}$$

Factor 4 out of $\mathrm{R1}$:

$$= 4\begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ -1 & 3 & 3 \\ 0 & -3 & -3 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + \mathrm{R1}$:

$$= 4\begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ 0 & \frac{13}{4} & \frac{9}{2} \\ 0 & -3 & -3 \end{vmatrix}$$

Factor $\frac{13}{4}$ out of $\mathrm{R2}$:

$$= 13\begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ 0 & 1 & \frac{18}{13} \\ 0 & -3 & -3 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + 3\mathrm{R2}$:

$$= 13\begin{vmatrix} 1 & \frac{1}{4} & \frac{3}{2} \\ 0 & 1 & \frac{18}{13} \\ 0 & 0 & \frac{15}{13} \end{vmatrix}$$

Then:

$$= 13 \times \left(1 \times 1 \times \frac{15}{13}\right)$$

$$= 15$$

9) d)

$$\begin{vmatrix} 3 & -2 & 5 \\ -2 & 4 & 2 \\ -4 & -1 & 9 \end{vmatrix}$$

Factor 3 out of R1:

$$= 3\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ -2 & 4 & 2 \\ -4 & -1 & 9 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + 2\mathrm{R1}$:

$$= 3\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & \frac{8}{3} & \frac{16}{3} \\ -4 & -1 & 9 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + 4\mathrm{R1}$:

$$= 3\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & \frac{8}{3} & \frac{16}{3} \\ 0 & -\frac{11}{3} & \frac{47}{3} \end{vmatrix}$$

Factor $\frac{8}{3}$ out of R2:

$$= 8\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & 1 & 2 \\ 0 & -\frac{11}{3} & \frac{47}{3} \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \frac{11}{3}\mathrm{R2}$:

$$= 8\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3} \\ 0 & 1 & 2 \\ 0 & 0 & 23 \end{vmatrix}$$

Then:

$$= 8 \times (1 \times 1 \times 23)$$

$$= 184$$

10) a)

$$|\mathbf{A}| = \begin{vmatrix} 2 & 1 & 3 \\ 0 & -2 & 3 \\ 0 & 0 & 5 \end{vmatrix} = 2\big((-2)\times 5 - 3\times 0\big) - (0\times 5 - 3\times 0) + 3(0\times 0 - (-2)\times 0)$$

$$= 2\times(-2)\times 5$$

$$= -20$$

10) b)

$$\begin{vmatrix} 0 & 0 & 5 \\ 2 & 1 & 3 \\ 0 & -2 & 3 \end{vmatrix} = 0\big(1\times 3 - 3\times(-2)\big) - 0(2\times 3 - 3\times 0) + 5(2\times(-2) - 1\times 0)$$

$$= 5\times 2\times(-2)$$

$$= -20$$

10) c)

$$\begin{vmatrix} 0 & -2 & 3 \\ 0 & 0 & 5 \\ 2 & 1 & 3 \end{vmatrix} = 0(0\times 3 - 5\times 1) + 2(0\times 3 - 5\times 2) + 3(0\times 1 - 0\times 2)$$

$$= 2\times(-5\times 2)$$

$$= -20$$

10) d)

$$\begin{vmatrix} 1 & 3 & 2 \\ -2 & 3 & 0 \\ 0 & 5 & 0 \end{vmatrix} = (3\times 0 - 0\times 5) - 3\big((-2)\times 0 - 0\times 0\big) + 2\big((-2)\times 5 - 3\times 0\big)$$

$$= 2\times(-2)\times 5$$

$$= -20$$

10) e)

$$\begin{vmatrix} 3 & 2 & 1 \\ 3 & 0 & -2 \\ 5 & 0 & 0 \end{vmatrix} = 3(0\times 0 - (-2)\times 0) - 2(3\times 0 - (-2)\times 5) + (2\times 0 - 0\times 5)$$

$$= -2\times(-(-2)\times 5)$$

$$= -20$$

10) f)

$$\begin{vmatrix} 0 & -2 & 3 \\ 2 & 1 & 3 \\ 0 & 0 & 5 \end{vmatrix} = 0(1 \times 5 - 3 \times 0) + 2(2 \times 5 - 3 \times 0) + 3(2 \times 0 - 1 \times 0)$$

$$= 2 \times 2 \times 5$$

$$= 20$$

10) g)

$$\begin{vmatrix} 1 & 2 & 3 \\ -2 & 0 & 3 \\ 0 & 0 & 5 \end{vmatrix} = (0 \times 5 - 3 \times 0) - 2\big((-2) \times 5 - 3 \times 0\big) + 3\big((-2) \times 0 - 0 \times 0\big)$$

$$= -2 \times (-2) \times 5$$

$$= 20$$

10) h)

If you rotate the rows or columns, the determinant does not change. If you swap any two rows or columns, then the determinant changes sign.

11)

R1 and R3 have been swapped, so the value is -6

12) a)

$$\begin{vmatrix} 3 & 0 & 0 \\ 0 & 0 & 2 \\ 0 & 5 & 0 \end{vmatrix}$$

Swap R2 and R3:

$$= -\begin{vmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

$$= -3 \times 5 \times 2$$

$$= -30$$

12) b)

$$\begin{vmatrix} 0 & 0 & 2 \\ 0 & 4 & 0 \\ -5 & 0 & 0 \end{vmatrix}$$

Swap R1 and R3:

$$= -\begin{vmatrix} -5 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

$$= -(-5) \times 4 \times 2$$

$$= 40$$

12) c)

$$\begin{vmatrix} 0 & 0 & 6k \\ k & 0 & 0 \\ 0 & 4k & 0 \end{vmatrix}$$

Rotate all rows up one position:

$$= \begin{vmatrix} k & 0 & 0 \\ 0 & 4k & 0 \\ 0 & 0 & 6k \end{vmatrix}$$

$$= k \times 4k \times 6k$$

$$= 24k^3$$

12) d)

$$\begin{vmatrix} 0 & pq & 0 \\ p+q & 0 & 0 \\ 0 & 0 & pq \end{vmatrix}$$

Swap C1 and C2:

$$= -\begin{vmatrix} pq & 0 & 0 \\ 0 & p+q & 0 \\ 0 & 0 & pq \end{vmatrix}$$

$$= -pq \times (p+q) \times pq$$

$$= -p^2q^2(p+q)$$

13) a)

$$\begin{vmatrix} 1 & 1 & 1 \\ 3 & 4 & -5 \\ 1 & 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R3}$:

$$= \begin{vmatrix} 0 & 0 & 0 \\ 3 & 4 & -5 \\ 1 & 1 & 1 \end{vmatrix} = 0$$ as all of the entries in a single row are zero.

13) b)

$$\begin{vmatrix} 2 & -3 & 4 \\ 3 & 0 & 6 \\ 4 & 2 & 8 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \frac{1}{2}\mathrm{C3}$:

$$= \begin{vmatrix} 0 & -3 & 4 \\ 0 & 0 & 6 \\ 0 & 2 & 8 \end{vmatrix} = 0$$ as all of the entries in a single column are zero.

14) a)

$$\begin{vmatrix} 4k+1 & k & 4k+1 \\ 3k & k & 3k \\ -2 & 4 & 2k+2 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$$= \begin{vmatrix} k+1 & 0 & k+1 \\ 3k & k & 3k \\ -2 & 4 & 2k+2 \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \mathrm{C1}$:

$$= \begin{vmatrix} k+1 & 0 & 0 \\ 3k & k & 0 \\ -2 & 4 & 2k+4 \end{vmatrix}$$

14) b)

$|\mathbf{A}| = (k+1) \times k \times (2k+4)$

$= 2k(k+1)(k+2)$

15) a)

$$\begin{vmatrix} 1 & p & -p \\ 1 & p+2 & 1 \\ -1 & -p & p+1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \mathrm{R1}$:

$$= \begin{vmatrix} 1 & p & -p \\ 1 & p+2 & 1 \\ 0 & 0 & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$= \begin{vmatrix} 1 & p & -p \\ 0 & 2 & 1+p \\ 0 & 0 & 1 \end{vmatrix}$$

Then:

$$= 1 \times 2 \times 1$$

$$= 2$$

15) b)

$$\begin{vmatrix} p-1 & 1 & p-1 \\ p-2 & 1 & p-2 \\ 3p-3 & p & 3p \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$$= \begin{vmatrix} 1 & 0 & 1 \\ p-2 & 1 & p-2 \\ 3p-3 & p & 3p \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \mathrm{C1}$:

$$= \begin{vmatrix} 1 & 0 & 0 \\ p-2 & 1 & 0 \\ 3p-3 & p & 3 \end{vmatrix}$$

Then:

$$= 1 \times 1 \times 3$$

$$= 3$$

15) c)

$$\begin{vmatrix} 2 & -p & -1 \\ 1 & 2p-5 & 2 \\ p & 2 & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + 2\mathrm{R1}$:

$$= \begin{vmatrix} 2 & -p & -1 \\ 5 & -5 & 0 \\ p & 2 & 1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \mathrm{R1}$:

$$= \begin{vmatrix} 2 & -p & -1 \\ 5 & -5 & 0 \\ p+2 & 2-p & 0 \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} + \mathrm{C1}$:

$$= \begin{vmatrix} 2 & 2-p & -1 \\ 5 & 0 & 0 \\ p+2 & 4 & 0 \end{vmatrix}$$

Then:

$= (-1) \times (5 \times 4)$

$= -20$

15) d)

$$\begin{vmatrix} p-2 & 3p-3 & p-1 \\ p+3 & p+14 & p+3 \\ p-1 & 3p-2 & p \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= \begin{vmatrix} p-2 & 3p-3 & p-1 \\ p+3 & p+14 & p+3 \\ 1 & 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C1}$:

$$= \begin{vmatrix} p-2 & 2p-1 & p-1 \\ p+3 & 11 & p+3 \\ 1 & 0 & 1 \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \mathrm{C1}$:

$$= \begin{vmatrix} p-2 & 2p-1 & 1 \\ p+3 & 11 & 0 \\ 1 & 0 & 0 \end{vmatrix}$$

Then:

$= 1 \times -1 \times 11$

$= -11$

16) a)

$$\begin{vmatrix} n+5 & 2n+4 & 9-n \\ 4-n & 4-n & 8-3n \\ n+3 & 2n+2 & 5 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C2}$:

$$= \begin{vmatrix} 1-n & 2n+4 & 9-n \\ 0 & 4-n & 8-3n \\ 1-n & 2n+2 & 5 \end{vmatrix}$$

Factor $(1-n)$ out of $\mathrm{C1}$:

$$= (1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 4-n & 8-3n \\ 1 & 2n+2 & 5 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= (1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 4-n & 8-3n \\ 0 & -2 & n-4 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + 2\mathrm{R3}$:

$$= (1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & -n & -n \\ 0 & -2 & n-4 \end{vmatrix}$$

Factor $(-n)$ out of $\mathrm{R2}$:

$$= -n(1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 1 & 1 \\ 0 & -2 & n-4 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + 2\mathrm{R2}$:

$$= -n(1-n)\begin{vmatrix} 1 & 2n+4 & 9-n \\ 0 & 1 & 1 \\ 0 & 0 & n-2 \end{vmatrix}$$

Then:

$$= -n(1-n)(n-2)$$

$$= n(n-1)(n-2)$$

16) b)

$$\begin{vmatrix} -2n & 1-3n & 1 \\ -4-n & 5-2n & 5 \\ -1-3n & 2-2n & n+2 \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C3}$:

$$= \begin{vmatrix} -2n & -3n & 1 \\ -4-n & -2n & 5 \\ -1-3n & -3n & n+2 \end{vmatrix}$$

Factor n out of C2:

$$= n\begin{vmatrix} -2n & -3 & 1 \\ -4-n & -2 & 5 \\ -1-3n & -3 & n+2 \end{vmatrix}$$

Rotate columns one to the left:

$$= n\begin{vmatrix} -3 & 1 & -2n \\ -2 & 5 & -4-n \\ -3 & n+2 & -1-3n \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \mathrm{C1}$:

$$= n\begin{vmatrix} -3 & 1 & -2n \\ -2 & 5 & -4-n \\ 0 & n+1 & -1-n \end{vmatrix} = n\begin{vmatrix} -3 & 1 & -2n \\ -2 & 5 & -4-n \\ 0 & n+1 & -(n+1) \end{vmatrix}$$

Factor $(n+1)$ out of R3:

$$= n(n+1)\begin{vmatrix} -3 & 1 & -2n \\ -2 & 5 & -4-n \\ 0 & 1 & -1 \end{vmatrix}$$

Multiply R1 by 2:

$$= \frac{1}{2}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ -2 & 5 & -4-n \\ 0 & 1 & -1 \end{vmatrix}$$

Multiply R2 by 3:

$$= \frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ -6 & 15 & -12-3n \\ 0 & 1 & -1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$= \frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ 0 & 13 & n-12 \\ 0 & 1 & -1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 13\mathrm{R3}$:

$$= \frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ 0 & 0 & n+1 \\ 0 & 1 & -1 \end{vmatrix}$$

Swap R2 and R3:

$$= -\frac{1}{6}n(n+1)\begin{vmatrix} -6 & 2 & -4n \\ 0 & 1 & -1 \\ 0 & 0 & n+1 \end{vmatrix}$$

Then:

$$= -\frac{1}{6}n(n+1) \times \big((-6) \times 1 \times (n+1)\big)$$

$$= n(n+1)^2$$

16) c)

$$\begin{vmatrix} 2n-4 & 3n-6 & 4-2n \\ -6 & 4n-6 & 2-5n \\ 3n-6 & 3n-6 & 2-n \end{vmatrix} = \begin{vmatrix} 2(n-2) & 3(n-2) & -2(n-2) \\ -6 & 4n-6 & 2-5n \\ 3(n-2) & 3(n-2) & -(n-2) \end{vmatrix}$$

Factor $(n-2)$ out of R1:

$$= (n-2)\begin{vmatrix} 2 & 3 & -2 \\ -6 & 4n-6 & 2-5n \\ 3(n-2) & 3(n-2) & -(n-2) \end{vmatrix}$$

Factor $(n-2)$ out of R3:

$$= (n-2)^2\begin{vmatrix} 2 & 3 & -2 \\ -6 & 4n-6 & 2-5n \\ 3 & 3 & -1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + 2\mathrm{R3}$:

$$= (n-2)^2\begin{vmatrix} 2 & 3 & -2 \\ 0 & 4n & -5n \\ 3 & 3 & -1 \end{vmatrix}$$

Factor n out of R2:

$$= n(n-2)^2\begin{vmatrix} 2 & 3 & -2 \\ 0 & 4 & -5 \\ 3 & 3 & -1 \end{vmatrix}$$

Factor 2 out of R1:

$$= 2n(n-2)^2\begin{vmatrix} 1 & \frac{3}{2} & -1 \\ 0 & 4 & -5 \\ 3 & 3 & -1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 3\mathrm{R1}$:

$$= 2n(n-2)^2\begin{vmatrix} 1 & \frac{3}{2} & -1 \\ 0 & 4 & -5 \\ 0 & -\frac{3}{2} & 2 \end{vmatrix}$$

Factor 4 out of R2:

$$= 8n(n-2)^2 \begin{vmatrix} 1 & \frac{3}{2} & -1 \\ 0 & 1 & -\frac{5}{4} \\ 0 & -\frac{3}{2} & 2 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \frac{3}{2}\mathrm{R2}$:

$$= 8n(n-2)^2 \begin{vmatrix} 1 & \frac{3}{2} & -1 \\ 0 & 1 & -\frac{5}{4} \\ 0 & 0 & \frac{1}{8} \end{vmatrix}$$

Then:

$= 8n(n-2)^2 \times \frac{1}{8}$

$= n(n-2)^2$

16) d)

$$\begin{vmatrix} 9-15n & 12-17n & 12-15n \\ 40-49n & 40-49n & 40-50n \\ 13-20n & 16-22n & 16-20n \end{vmatrix} = \begin{vmatrix} 9-15n & 12-17n & 3(4-5n) \\ 40-49n & 40-49n & 10(4-5n) \\ 13-20n & 16-22n & 4(4-5n) \end{vmatrix}$$

Factor $(4-5n)$ out of C3:

$$= (4-5n)\begin{vmatrix} 9-15n & 12-17n & 3 \\ 40-49n & 40-49n & 10 \\ 13-20n & 16-22n & 4 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C2}$:

$$= (4-5n)\begin{vmatrix} 2n-3 & 12-17n & 3 \\ 0 & 40-49n & 10 \\ 2n-3 & 16-22n & 4 \end{vmatrix}$$

Factor $(2n-3)$ out of C1:

$$= (2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n & 3 \\ 0 & 40-49n & 10 \\ 1 & 16-22n & 4 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= (2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n & 3 \\ 0 & 40-49n & 10 \\ 0 & 4-5n & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 10\mathrm{R3}$:

$$= (2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n & 3 \\ 0 & n & 0 \\ 0 & 4-5n & 1 \end{vmatrix}$$

Factor n out of R2:

$$= n(2n-3)(4-5n)\begin{vmatrix} 1 & 12-17n & 3 \\ 0 & 1 & 0 \\ 0 & 4-5n & 1 \end{vmatrix}$$

Then:

$$= n(2n-3)(4-5n)$$

17) a)

At stage (3), the a_{33} element (bottom right), should be $4n-1$

as $(n-3)+(3n+2) = 4n-1$

17) b)

$$\begin{vmatrix} n-1 & 2 & -1 \\ -n & n & n \\ -2n-1 & 3n+2 & 4n+1 \end{vmatrix}$$

Factor n out of R2:

$$= n\begin{vmatrix} n-1 & 2 & -1 \\ -1 & 1 & 1 \\ -2n-1 & 3n+2 & 4n+1 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} + \mathrm{C2}$:

$$= n\begin{vmatrix} n+1 & 2 & -1 \\ 0 & 1 & 1 \\ n+1 & 3n+2 & 4n+1 \end{vmatrix}$$

Factor $(n+1)$ out of C1:

$$= n(n+1)\begin{vmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 1 & 3n+2 & 4n+1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= n(n+1)\begin{vmatrix} 1 & 2 & -1 \\ 0 & 1 & 1 \\ 0 & 3n & 4n+2 \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \mathrm{C2}$:

$$= n(n+1)\begin{vmatrix} 1 & 2 & -3 \\ 0 & 1 & 0 \\ 0 & 3n & n+2 \end{vmatrix}$$

Then:

$$= n(n+1)(n+2)$$

18) a)

$$\begin{vmatrix} n+1 & n^2 & 1 \\ -n-1 & n+1-n^2 & n \\ n+1 & 2n^2-n & 2n^2-n \end{vmatrix} = \begin{vmatrix} n+1 & n^2 & 1 \\ -(n+1) & n+1-n^2 & n \\ n+1 & 2n^2-n & 2n^2-n \end{vmatrix}$$

Factor $(n+1)$ out of C1:

$$= (n+1)\begin{vmatrix} 1 & n^2 & 1 \\ -1 & n+1-n^2 & n \\ 1 & 2n^2-n & 2n^2-n \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= (n+1)\begin{vmatrix} 1 & n^2 & 1 \\ -1 & n+1-n^2 & n \\ 0 & n^2-n & 2n^2-n-1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + \mathrm{R1}$:

$$= (n+1)\begin{vmatrix} 1 & n^2 & 1 \\ 0 & n+1 & n+1 \\ 0 & n^2-n & 2n^2-n-1 \end{vmatrix}$$

Factor $(n+1)$ out of R2:

$$= (n+1)^2\begin{vmatrix} 1 & n^2 & 1 \\ 0 & 1 & 1 \\ 0 & n^2-n & 2n^2-n-1 \end{vmatrix} = (n+1)^2\begin{vmatrix} 1 & n^2 & 1 \\ 0 & 1 & 1 \\ 0 & n(n-1) & (2n+1)(n-1) \end{vmatrix}$$

Factor $(n-1)$ out of R3:

$$= (n+1)^2(n-1)\begin{vmatrix} 1 & n^2 & 1 \\ 0 & 1 & 1 \\ 0 & n & 2n+1 \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \mathrm{C2}$:

$$= (n+1)^2(n-1)\begin{vmatrix} 1 & n^2 & 1-n^2 \\ 0 & 1 & 0 \\ 0 & n & n+1 \end{vmatrix} = (n+1)^2(n-1)\begin{vmatrix} 1 & n^2 & (1-n)(n+1) \\ 0 & 1 & 0 \\ 0 & n & n+1 \end{vmatrix}$$

Factor $(n+1)$ out of C3:

$$= (n+1)^3(n-1)\begin{vmatrix} 1 & n^2 & 1-n \\ 0 & 1 & 0 \\ 0 & n & 1 \end{vmatrix}$$

Then:

$$= (n+1)^3(n-1)$$

18) b)

$$\begin{vmatrix} 3n-2 & n^2+2n & 2n-1 \\ 3n^2-1 & n+2 & 3n^2-1 \\ n^2+4n-4 & n^2+2n & n^2+3n-3 \end{vmatrix} = \begin{vmatrix} 3n-2 & n(n+2) & 2n-1 \\ 3n^2-1 & n+2 & 3n^2-1 \\ n^2+4n-4 & n(n+2) & n^2+3n-3 \end{vmatrix}$$

Factor $(n+2)$ out of C2:

$$= (n+2)\begin{vmatrix} 3n-2 & n & 2n-1 \\ 3n^2-1 & 1 & 3n^2-1 \\ n^2+4n-4 & n & n^2+3n-3 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C3}$:

$$= (n+2)\begin{vmatrix} n-1 & n & 2n-1 \\ 0 & 1 & 3n^2-1 \\ n-1 & n & n^2+3n-3 \end{vmatrix}$$

Factor $(n-1)$ out of C1:

$$= (n-1)(n+2)\begin{vmatrix} 1 & n & 2n-1 \\ 0 & 1 & 3n^2-1 \\ 1 & n & n^2+3n-3 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= (n-1)(n+2)\begin{vmatrix} 1 & n & 2n-1 \\ 0 & 1 & 3n^2-1 \\ 0 & 0 & n^2+n-2 \end{vmatrix} = (n-1)(n+2)\begin{vmatrix} 1 & n & 2n-1 \\ 0 & 1 & 3n^2-1 \\ 0 & 0 & (n+2)(n-1) \end{vmatrix}$$

Factor $(n+2)(n-1)$ out of R3:

$$= (n-1)^2(n+2)^2\begin{vmatrix} 1 & n & 2n-1 \\ 0 & 1 & 3n^2-1 \\ 0 & 0 & 1 \end{vmatrix}$$

Then:

$$= (n-1)^2(n+2)^2$$

18) c)

$$\begin{vmatrix} 3n-n^2 & n+1 & n^2-n-6 \\ 6+n-n^2 & -n^2-n-2 & n^2-9 \\ 6-2n & -n^2-3n & n-3 \end{vmatrix}$$

$$= \begin{vmatrix} n(3-n) & n+1 & (n-3)(n+2) \\ (n+2)(3-n) & -n^2-n-2 & (n-3)(n+3) \\ 2(3-n) & -n^2-3n & n-3 \end{vmatrix}$$

Factor $(3-n)$ out of C1:

$$= (3-n)\begin{vmatrix} n & n+1 & (n-3)(n+2) \\ n+2 & -n^2-n-2 & (n-3)(n+3) \\ 2 & -n^2-3n & n-3 \end{vmatrix}$$

Factor $(n-3)$ out of C3:

$$= (3-n)(n-3)\begin{vmatrix} n & n+1 & n+2 \\ n+2 & -n^2-n-2 & n+3 \\ 2 & -n^2-3n & 1 \end{vmatrix}$$

$$= -(n-3)^2\begin{vmatrix} n & n+1 & n+2 \\ n+2 & -n^2-n-2 & n+3 \\ 2 & -n^2-3n & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$= -(n-3)^2\begin{vmatrix} n & n+1 & n+2 \\ 2 & -n^2-2n-3 & 1 \\ 2 & -n^2-3n & 1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R2}$:

$$= -(n-3)^2\begin{vmatrix} n & n+1 & n+2 \\ 2 & -n^2-2n-3 & 1 \\ 0 & 3-n & 0 \end{vmatrix}$$

Factor $(3-n)$ out of R3:

$$= -(n-3)^2(3-n)\begin{vmatrix} n & n+1 & n+2 \\ 2 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

$$= (n-3)^3\begin{vmatrix} n & n+1 & n+2 \\ 2 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C3}$:

$$= (n-3)^3\begin{vmatrix} -2 & n+1 & n+2 \\ 1 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + 2\mathrm{R2}$:

$$= (n-3)^3\begin{vmatrix} 0 & -2n^2-3n-5 & n+4 \\ 1 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

Rotate rows one place upwards:

$$= (n-3)^3 \begin{vmatrix} 1 & -n^2-2n-3 & 1 \\ 0 & 1 & 0 \\ 0 & -2n^2-3n-5 & n+4 \end{vmatrix}$$

Then:

$$= (n-3)^3(n+4)$$

18) d)

$$\begin{vmatrix} n-7 & n-12 & n-10 \\ 5n^2-20n & 11n^2-44n & 10n^2-40n \\ 3n-6 & 5n-10 & 5n-8 \end{vmatrix}$$

$$= \begin{vmatrix} n-7 & n-12 & n-10 \\ 5n(n-4) & 11n(n-4) & 10n(n-4) \\ 3n-6 & 5n-10 & 5n-8 \end{vmatrix}$$

Factor $n(n-4)$ out of R2:

$$= n(n-4)\begin{vmatrix} n-7 & n-12 & n-10 \\ 5 & 11 & 10 \\ 3n-6 & 5n-10 & 5n-8 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= n(n-4)\begin{vmatrix} n-7 & n-12 & n-10 \\ 5 & 11 & 10 \\ 2n+1 & 4n+2 & 4n+2 \end{vmatrix}$$

$$= n(n-4)\begin{vmatrix} n-7 & n-12 & n-10 \\ 5 & 11 & 10 \\ 2n+1 & 2(2n+1) & 2(2n+1) \end{vmatrix}$$

Factor $(2n+1)$ out of R3:

$$= n(n-4)(2n+1)\begin{vmatrix} n-7 & n-12 & n-10 \\ 5 & 11 & 10 \\ 1 & 2 & 2 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C2}$:

$$= n(n-4)(2n+1)\begin{vmatrix} 5 & n-12 & n-10 \\ -6 & 11 & 10 \\ -1 & 2 & 2 \end{vmatrix}$$

Factor 5 out of R1:

$$= 5n(n-4)(2n+1)\begin{vmatrix} 1 & \frac{1}{5}n-\frac{12}{5} & \frac{1}{5}n-2 \\ -6 & 11 & 10 \\ -1 & 2 & 2 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + 6\mathrm{R1}$:

$$= 5n(n-4)(2n+1)\begin{vmatrix} 1 & \frac{1}{5}n-\frac{12}{5} & \frac{1}{5}n-2 \\ 0 & \frac{6}{5}n-\frac{17}{5} & \frac{6}{5}n-2 \\ -1 & 2 & 2 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \mathrm{R1}$:

$$= 5n(n-4)(2n+1)\begin{vmatrix} 1 & \frac{1}{5}n-\frac{12}{5} & \frac{1}{5}n-2 \\ 0 & \frac{6}{5}n-\frac{17}{5} & \frac{6}{5}n-2 \\ 0 & \frac{1}{5}n-\frac{2}{5} & \frac{1}{5}n \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C3}$:

$$= 5n(n-4)(2n+1)\begin{vmatrix} 1 & -\frac{2}{5} & \frac{1}{5}n-2 \\ 0 & -\frac{7}{5} & \frac{6}{5}n-2 \\ 0 & -\frac{2}{5} & \frac{1}{5}n \end{vmatrix}$$

Factor $-\frac{7}{5}$ out of R2:

$$= -7n(n-4)(2n+1)\begin{vmatrix} 1 & -\frac{2}{5} & \frac{1}{5}n-2 \\ 0 & 1 & \frac{10}{7}-\frac{6}{7}n \\ 0 & -\frac{2}{5} & \frac{1}{5}n \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \frac{2}{5}\mathrm{R2}$:

$$= -7n(n-4)(2n+1)\begin{vmatrix} 1 & -\frac{2}{5} & \frac{1}{5}n-2 \\ 0 & 1 & \frac{10}{7}-\frac{6}{7}n \\ 0 & -0 & \frac{4}{7}-\frac{n}{7} \end{vmatrix}$$

$$= -7n(n-4)(2n+1)\begin{vmatrix} 1 & -\frac{2}{5} & \frac{1}{5}n-2 \\ 0 & 1 & \frac{10}{7}-\frac{6}{7}n \\ 0 & -0 & -\frac{1}{7}(n-4) \end{vmatrix}$$

Then:

$$= -7n(n-4)(2n+1) \times -\frac{1}{7}(n-4)$$

$$= n(n-4)^2(2n+1)$$

19)

$$|\mathbf{A}| = \begin{vmatrix} 5p+q & -2q & 3p-3q \\ 9p & 9q & 6p-6q \\ 3pq+3q^2 & -3q^2 & pq-q^2 \end{vmatrix}$$

Factor q out of C2:

$$= q\begin{vmatrix} 5p+q & -2 & 3p-3q \\ 9p & 9 & 6p-6q \\ 3pq+3q^2 & -3q & pq-q^2 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C3}$:

$$= q\begin{vmatrix} 2p+4q & -2 & 3p-3q \\ 3p+6q & 9 & 6p-6q \\ 2pq+4q^2 & -3q & pq-q^2 \end{vmatrix}$$

$$= q\begin{vmatrix} 2(p+2q) & -2 & 3(p-q) \\ 3(p+2q) & 9 & 6(p-q) \\ 2q(p+2q) & -3q & q(p-q) \end{vmatrix}$$

Factor $(p+2q)$ out of C1:

$$= q(p+2q)\begin{vmatrix} 2 & -2 & 3(p-q) \\ 3 & 9 & 6(p-q) \\ 2q & -3q & q(p-q) \end{vmatrix}$$

Factor $(p-q)$ out of C3:

$$= q(p+2q)(p-q)\begin{vmatrix} 2 & -2 & 3 \\ 3 & 9 & 6 \\ 2q & -3q & q \end{vmatrix}$$

Factor q out of R3:

$$= q^2(p+2q)(p-q)\begin{vmatrix} 2 & -2 & 3 \\ 3 & 9 & 6 \\ 2 & -3 & 1 \end{vmatrix}$$

$$= q^2(p+2q)(p-q)\big(2\big(9\times 1 - 6\times(-3)\big) + 2(3\times 1 - 6\times 2) + 3(3\times(-3) - 9\times 2)\big)$$

$$= q^2(p+2q)(p-q)\big(2\times 27 + 2\times(-9) + 3\times(-27)\big)$$

$$= -45q^2(p+2q)(p-q)$$

$$= 45q^2(p+2q)(q-p)$$

20) a)

$$\begin{vmatrix} 1 & p & pqr \\ 2 & q & pqr \\ 3 & r & pqr \end{vmatrix}$$

Factor pqr out of C3:

$$= pqr \begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 3 & r & 1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 3\mathrm{R1}$:

$$= pqr \begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 0 & r-3p & -2 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R2} - 2\mathrm{R1}$:

$$= pqr \begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 0 & r-3p & -2 \end{vmatrix}$$

Multiply R2 by 2:

$$= \frac{1}{2} pqr \begin{vmatrix} 1 & p & 1 \\ 0 & 2q-4p & -2 \\ 0 & r-3p & -2 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R3}$:

$$= \frac{1}{2} pqr \begin{vmatrix} 1 & p & 1 \\ 0 & 2q-p-r & 0 \\ 0 & r-3p & -2 \end{vmatrix}$$

Then:

$$= \frac{1}{2} pqr \times \big(1 \times (2q-p-r) \times (-2)\big)$$

$$= pqr(p-2q+r)$$

20) b)

$$\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \mathrm{R2}$:

$$= \begin{vmatrix} 1 & p & p+q+r \\ 2 & q & p+q+r \\ 3 & r & p+q+r \end{vmatrix}$$

Factor $(p+q+r)$ out of C3:

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 3 & r & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R1}$:

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 3 & r & 1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 3\mathrm{R1}$:

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 0 & r-3p & -2 \end{vmatrix}$$

Then:

$$= (p+q+r)\big((q-2p)\times(-2) - (-1)\times(r-3p)\big)$$

$$= (p+q+r)(-2q+4p+r-3p)$$

$$= (p+q+r)(p-2q+r)$$

20) c)

$$\begin{vmatrix} 1 & p & pr \\ 2 & q & qr \\ 3 & r & pq \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R1}$:

$$= \begin{vmatrix} 1 & p & pr \\ 0 & q-2p & qr-2pr \\ 3 & r & pq \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 3\mathrm{R1}$:

$$= \begin{vmatrix} 1 & p & pr \\ 0 & q-2p & qr-2pr \\ 0 & r-3p & pq-3pr \end{vmatrix} = \begin{vmatrix} 1 & p & pr \\ 0 & q-2p & r(q-2p) \\ 0 & r-3p & pq-3pr \end{vmatrix}$$

Factor $(q-2p)$ out of R2:

$$= (q-2p)\begin{vmatrix} 1 & p & pr \\ 0 & 1 & r \\ 0 & r-3p & pq-3pr \end{vmatrix}$$

Then:

$$= (q-2p)\big((pq-3pr) - r(r-3p)\big)$$
$$= (q-2p)(pq-3pr-r^2+3pr)$$
$$= (q-2p)(pq-r^2)$$

21) a)

$$\begin{vmatrix} p-q-r & p-q-r & p-q-r \\ p-q-r & 2p-q & p+3q-r \\ p-q-r & p-q & p-q \end{vmatrix}$$

Factor $(p-q-r)$ out of C1:

$$= (p-q-r)\begin{vmatrix} 1 & p-q-r & p-q-r \\ 1 & 2p-q & p+3q-r \\ 1 & p-q & p-q \end{vmatrix}$$

Let the new $C3 = C3 - C2$:

$$= (p-q-r)\begin{vmatrix} 1 & p-q-r & 0 \\ 1 & 2p-q & -p+4q-r \\ 1 & p-q & 0 \end{vmatrix}$$

Factor $(-p+4q-r)$ out of C3:

$$= (p-q-r)(-p+4q-r)\begin{vmatrix} 1 & p-q-r & 0 \\ 1 & 2p-q & 1 \\ 1 & p-q & 0 \end{vmatrix}$$

Swap C1 and C3:

$$= -(p-q-r)(-p+4q-r)\begin{vmatrix} 0 & p-q-r & 1 \\ 1 & 2p-q & 1 \\ 0 & p-q & 1 \end{vmatrix}$$

Swap R1 and R2:

$$= (p-q-r)(-p+4q-r)\begin{vmatrix} 1 & 2p-q & 1 \\ 0 & p-q-r & 1 \\ 0 & p-q & 1 \end{vmatrix}$$

Then:

$$= (p-q-r)(-p+4q-r)\big((p-q-r)\times 1 - 1\times(p-q)\big)$$
$$= (p-q-r)(-p+4q-r)(p-q-r-p+q)$$
$$= -r(p-q-r)(-p+4q-r)$$
$$= r(p-q-r)(p-4q+r)$$

21) b)

$$\begin{vmatrix} -5q-7r & -3p-3q+3r & 2p-2q-4r \\ 3p-q-3r & p+q-r & 2p-2q-4r \\ p-6q-9r & -3p-3q+3r & 3p-3q-6r \end{vmatrix}$$

$$= \begin{vmatrix} -5q-7r & -3(p+q-r) & 2(p-q-2r) \\ 3p-q-3r & p+q-r & 2(p-q-2r) \\ p-6q-9r & -3(p+q-r) & 3(p-q-2r) \end{vmatrix}$$

Factor $(p+q-r)$ out of C2:

$$= (p+q-r)\begin{vmatrix} -5q-7r & -3 & 2(p-q-2r) \\ 3p-q-3r & 1 & 2(p-q-2r) \\ p-6q-9r & -3 & 3(p-q-2r) \end{vmatrix}$$

Factor $(p-q-2r)$ out of C3:

$$= (p+q-r)(p-q-2r)\begin{vmatrix} -5q-7r & -3 & 2 \\ 3p-q-3r & 1 & 2 \\ p-6q-9r & -3 & 3 \end{vmatrix}$$

Rotate columns one step to the right:

$$= (p+q-r)(p-q-2r)\begin{vmatrix} -3 & 2 & -5q-7r \\ 1 & 2 & 3p-q-3r \\ -3 & 3 & p-6q-9r \end{vmatrix}$$

Factor -3 out of R1:

$$= -3(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 1 & 2 & 3p-q-3r \\ -3 & 3 & p-6q-9r \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$= -3(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & \frac{8}{3} & 3p-\frac{8}{3}q-\frac{16}{3}r \\ -3 & 3 & p-6q-9r \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + 3\mathrm{R1}$:

$$= -3(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & \frac{8}{3} & 3p-\frac{8}{3}q-\frac{16}{3}r \\ 0 & 1 & p-q-2r \end{vmatrix}$$

Factor $\frac{8}{3}$ out of R2:

$$= -8(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & 1 & \frac{9}{8}p-q-2r \\ 0 & 1 & p-q-2r \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R2}$:

$$= -8(p+q-r)(p-q-2r)\begin{vmatrix} 1 & -\frac{2}{3} & \frac{5}{3}q+\frac{7}{3}r \\ 0 & 1 & \frac{9}{8}p-q-2r \\ 0 & 0 & -\frac{1}{8}p \end{vmatrix}$$

Then:

$$= -8(p+q-r)(p-q-2r) \times -\frac{1}{8}p$$

$$= p(p+q-r)(p-q-2r)$$

21) c)

$$\begin{vmatrix} 2p+q+r & p & p+r \\ -2p-2r & -p+q-r & -2p-2r \\ 3p+q+2r & 2p+r & 2p+2r \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= \begin{vmatrix} 2p+q+r & p & p+r \\ -2p-2r & -p+q-r & -2p-2r \\ p+r & p+r & p+r \end{vmatrix}$$

Factor $(p+r)$ out of R3:

$$= (p+r)\begin{vmatrix} 2p+q+r & p & p+r \\ -2p-2r & -p+q-r & -2p-2r \\ 1 & 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C2}$:

$$= (p+r)\begin{vmatrix} p+q+r & p & p+r \\ -p-q-r & -p+q-r & -2p-2r \\ 0 & 1 & 1 \end{vmatrix}$$

$$= (p+r)\begin{vmatrix} p+q+r & p & p+r \\ -(p+q+r) & -p+q-r & -2p-2r \\ 0 & 1 & 1 \end{vmatrix}$$

Factor $(p+q+r)$ out of C1:

$$= (p+r)(p+q+r)\begin{vmatrix} 1 & p & p+r \\ -1 & -p+q-r & -2p-2r \\ 0 & 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + \mathrm{R1}$:

$$= (p+r)(p+q+r)\begin{vmatrix} 1 & p & p+r \\ 0 & q-r & -p-r \\ 0 & 1 & 1 \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - \mathrm{C3}$:

$$= (p+r)(p+q+r)\begin{vmatrix} 1 & -r & p+r \\ 0 & p+q & -p-r \\ 0 & 0 & 1 \end{vmatrix}$$

Then:

$$= (p+r)(p+q+r) \times (p+q)$$

$$= (p+q)(p+r)(p+q+r)$$

21) d)

$$\begin{vmatrix} 2p-4q+r & 3p-3q & 3p-6q+3r \\ p-q & p-q & p-3q+2r \\ p-3q+r & 2p-2q & 2p-2q \end{vmatrix}$$

$$= \begin{vmatrix} 2p-4q+r & 3(p-q) & 3p-6q+3r \\ p-q & p-q & p-3q+2r \\ p-3q+r & 2(p-q) & 2p-2q \end{vmatrix}$$

Factor $(p-q)$ out of C2:

$$= (p-q)\begin{vmatrix} 2p-4q+r & 3 & 3p-6q+3r \\ p-q & 1 & p-3q+2r \\ p-3q+r & 2 & 2p-2q \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - 3\mathrm{R2}$:

$$= (p-q)\begin{vmatrix} -p-q+r & 0 & 3q-3r \\ p-q & 1 & p-3q+2r \\ p-3q+r & 2 & 2p-2q \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 2\mathrm{R2}$:

$$= (p-q)\begin{vmatrix} -p-q+r & 0 & 3q-3r \\ p-q & 1 & p-3q+2r \\ -p-q+r & 0 & 4q-4r \end{vmatrix}$$

Swap R1 and R2:

$$= -(p-q)\begin{vmatrix} p-q & 1 & p-3q+2r \\ -p-q+r & 0 & 3q-3r \\ -p-q+r & 0 & 4q-4r \end{vmatrix}$$

Swap C1 and C2:

$$= (p-q)\begin{vmatrix} 1 & p-q & p-3q+2r \\ 0 & -p-q+r & 3q-3r \\ 0 & -p-q+r & 4q-4r \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R2}$:

$$= (p-q)\begin{vmatrix} 1 & p-q & p-3q+2r \\ 0 & -p-q+r & 3q-3r \\ 0 & 0 & q-r \end{vmatrix}$$

Then:

$$= (p-q)(-p-q+r)(q-r)$$

22) a)

$$\begin{vmatrix} p & p^2 & p^3 \\ q & q^2 & q^3 \\ r & r^2 & r^3 \end{vmatrix}$$

Factor p out of R1:

$$= p\begin{vmatrix} 1 & p & p^2 \\ q & q^2 & q^3 \\ r & r^2 & r^3 \end{vmatrix}$$

Factor q out of R2:

$$= pq\begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ r & r^2 & r^3 \end{vmatrix}$$

Factor r out of R3:

$$= pqr\begin{vmatrix} 1 & p & p^2 \\ 1 & q & q^2 \\ 1 & r & r^2 \end{vmatrix}$$

Let the new $\text{R2} = \text{R2} - \text{R1}$:

$$= pqr\begin{vmatrix} 1 & p & p^2 \\ 0 & q-p & q^2-p^2 \\ 1 & r & r^2 \end{vmatrix}$$

Let the new $\text{R3} = \text{R3} - \text{R1}$:

$$= pqr\begin{vmatrix} 1 & p & p^2 \\ 0 & q-p & q^2-p^2 \\ 0 & r-p & r^2-p^2 \end{vmatrix} = pqr\begin{vmatrix} 1 & p & p^2 \\ 0 & q-p & (q-p)(q+p) \\ 0 & r-p & (r-p)(r+p) \end{vmatrix}$$

Factor $(q-p)$ out of R2:

$$= pqr(q-p)\begin{vmatrix} 1 & p & p^2 \\ 0 & 1 & q+p \\ 0 & r-p & (r-p)(r+p) \end{vmatrix}$$

Factor $(r-p)$ out of R2:

$$= pqr(q-p)(r-p)\begin{vmatrix} 1 & p & p^2 \\ 0 & 1 & q+p \\ 0 & 1 & r+p \end{vmatrix}$$

Let the new $\text{R3} = \text{R3} - \text{R2}$:

$$= pqr(q-p)(r-p)\begin{vmatrix} 1 & p & p^2 \\ 0 & 1 & q+p \\ 0 & 0 & r-q \end{vmatrix}$$

Then:

$= pqr(q-p)(r-p)(r-q)$

22) b)

$$\begin{vmatrix} 1 & p & p^3 \\ 1 & q & q^3 \\ 1 & r & r^3 \end{vmatrix}$$

Let the new R2 = R2 − R1:

$$= \begin{vmatrix} 1 & p & p^3 \\ 0 & q-p & q^3-p^3 \\ 1 & r & r^3 \end{vmatrix}$$

Let the new R3 = R3 − R1:

$$= \begin{vmatrix} 1 & p & p^3 \\ 0 & q-p & q^3-p^3 \\ 0 & r-p & r^3-p^3 \end{vmatrix} = \begin{vmatrix} 1 & p & p^3 \\ 0 & q-p & (q-p)(q^2+pq+p^2) \\ 0 & r-p & (r-p)(r^2+pr+p^2) \end{vmatrix}$$

Factor $(q-p)$ out of R2:

$$= (q-p)\begin{vmatrix} 1 & p & p^3 \\ 0 & 1 & q^2+pq+p^2 \\ 0 & r-p & (r-p)(r^2+pr+p^2) \end{vmatrix}$$

Factor $(r-p)$ out of R3:

$$= (q-p)(r-p)\begin{vmatrix} 1 & p & p^3 \\ 0 & 1 & q^2+pq+p^2 \\ 0 & 1 & r^2+pr+p^2 \end{vmatrix}$$

Let the new R3 = R3 − R2:

$$= (q-p)(r-p)\begin{vmatrix} 1 & p & p^3 \\ 0 & 1 & q^2+pq+p^2 \\ 0 & 0 & pr-pq+r^2-q^2 \end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix} 1 & p & p^3 \\ 0 & 1 & q^2+pq+p^2 \\ 0 & 0 & p(r-q)+(r-q)(r+q) \end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix} 1 & p & p^3 \\ 0 & 1 & q^2+pq+p^2 \\ 0 & 0 & (r-q)(p+q+r) \end{vmatrix}$$

Then:

$= (q-p)(r-p)(r-q)(p+q+r)$

22) c)

$$\begin{vmatrix} 1 & p^2 & p^3 \\ 1 & q^2 & q^3 \\ 1 & r^2 & r^3 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$= \begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q^2 - p^2 & q^3 - p^3 \\ 1 & r^2 & r^3 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R1}$:

$$= \begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q^2 - p^2 & q^3 - p^3 \\ 0 & r^2 - p^2 & r^3 - p^3 \end{vmatrix} = \begin{vmatrix} 1 & p^2 & p^3 \\ 0 & (q-p)(q+p) & (q-p)(q^2+pq+p^2) \\ 0 & (r-p)(r+p) & (r-p)(r^2+pr+p^2) \end{vmatrix}$$

Factor $(q-p)$ out of R2:

$$= (q-p)\begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q+p & q^2+pq+p^2 \\ 0 & (r-p)(r+p) & (r-p)(r^2+pr+p^2) \end{vmatrix}$$

Factor $(r-p)$ out of R3:

$$= (q-p)(r-p)\begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q+p & q^2+pq+p^2 \\ 0 & r+p & r^2+pr+p^2 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R2}$:

$$= (q-p)(r-p)\begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q+p & q^2+pq+p^2 \\ 0 & r-q & pr-pq+r^2-q^2 \end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q+p & q^2+pq+p^2 \\ 0 & r-q & p(r-q)+(r-q)(r+q) \end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q+p & q^2+pq+p^2 \\ 0 & r-q & (r-q)(p+q+r) \end{vmatrix}$$

Factor $(r-q)$ out of R3:

$$= (q-p)(r-p)(r-q)\begin{vmatrix} 1 & p^2 & p^3 \\ 0 & q+p & q^2+pq+p^2 \\ 0 & 1 & p+q+r \end{vmatrix}$$

Then:

$$= (q-p)(r-p)(r-q) \times \big((q+p) \times (p+q+r) - (q^2+pq+p^2) \times 1\big)$$

$= (q-p)(r-p)(r-q) \times (pq + q^2 + qr + p^2 + pq + pr - q^2 - pq - p^2)$
$= (q-p)(r-p)(r-q)(pq + pr + qr)$

22) d)

$$\begin{vmatrix} 1 & p & p^2(p+1) \\ 1 & q & q^2(q+1) \\ 1 & r & r^2(r+1) \end{vmatrix} = \begin{vmatrix} 1 & p & p^3+p^2 \\ 1 & q & q^3+q^2 \\ 1 & r & r^3+r^2 \end{vmatrix}$$

Let the new R2 $=$ R2 $-$ R1:

$$= \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & q-p & q^3-p^3+q^2-p^2 \\ 1 & r & r^3+r^2 \end{vmatrix}$$

Let the new R3 $=$ R3 $-$ R1:

$$= \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & q-p & q^3-p^3+q^2-p^2 \\ 0 & r-p & r^3-p^3+r^2-p^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & q-p & (q-p)(q^2+pq+p^2)+(q-p)(q+p) \\ 0 & r-p & (r-p)(r^2+pr+p^2)+(r-p)(r+p) \end{vmatrix}$$

Factor $(q-p)$ out of R2:

$$= (q-p)\begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & r-p & (r-p)(r^2+pr+p^2)+(r-p)(r+p) \end{vmatrix}$$

Factor $(r-p)$ out of R2:

$$= (q-p)(r-p)\begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 1 & r^2+pr+p^2+r+p \end{vmatrix}$$

Let the new R3 $=$ R3 $-$ R2:

$$= (q-p)(r-p)\begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 0 & r^2-q^2+pr-pq+r-q \end{vmatrix}$$

$$= (q-p)(r-p)\begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 0 & (r-q)(r+q)+p(r-q)+(r-q) \end{vmatrix}$$

Factor $(r-q)$ out of R3:

$$= (q-p)(r-p)(r-q)\begin{vmatrix} 1 & p & p^3+p^2 \\ 0 & 1 & q^2+pq+p^2+q+p \\ 0 & 0 & p+q+r+1 \end{vmatrix}$$

Then:

$$= (q-p)(r-p)(r-q)(p+q+r+1)$$

23) a)

$$\begin{vmatrix} 1 & p & q+r \\ 2 & q & p+r \\ 3 & r & p+q \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} + \mathrm{C2}$:

$$= \begin{vmatrix} 1 & p & p+q+r \\ 2 & q & p+q+r \\ 3 & r & p+q+r \end{vmatrix}$$

Factor $(p+q+r)$ out of C3:

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 2 & q & 1 \\ 3 & r & 1 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R1}$:

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 3 & r & 1 \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 3\mathrm{R1}$:

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 0 & q-2p & -1 \\ 0 & r-3p & -2 \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - \frac{1}{2}\mathrm{R3}$:

$$= (p+q+r)\begin{vmatrix} 1 & p & 1 \\ 0 & q-\frac{1}{2}p-\frac{1}{2}r & 0 \\ 0 & r-3p & -2 \end{vmatrix}$$

Then:

$$= (p+q+r)\left(1 \times \left(q-\frac{1}{2}p-\frac{1}{2}r\right) \times (-2)\right)$$

$$= (p+q+r)(p-2q+r)$$

23) b)

$$\begin{vmatrix} 1 & p+q & r \\ 2 & q+r & p \\ 4 & p+r & q \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} + \mathrm{C3}$:

$$= \begin{vmatrix} 1 & p+q+r & r \\ 2 & p+q+r & p \\ 4 & p+q+r & q \end{vmatrix}$$

Factor $(p+q+r)$ out of C2:

$$= (p+q+r)\begin{vmatrix} 1 & 1 & r \\ 2 & 1 & p \\ 4 & 1 & q \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R1}$:

$$= (p+q+r)\begin{vmatrix} 1 & 1 & r \\ 0 & -1 & p-2r \\ 4 & 1 & q \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 4\mathrm{R1}$:

$$= (p+q+r)\begin{vmatrix} 1 & 1 & r \\ 0 & -1 & p-2r \\ 0 & -3 & q-4r \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - 3\mathrm{R2}$:

$$= (p+q+r)\begin{vmatrix} 1 & 1 & r \\ 0 & -1 & p-2r \\ 0 & 0 & q+2r-3p \end{vmatrix}$$

Then:

$$= (p+q+r)\big(1 \times (-1) \times (q+2r-3p)\big)$$

$$= (p+q+r)(3p-q-2r)$$

23) c)

$$\begin{vmatrix} 1 & 3 & 6 \\ p+2r & q+2p & r+2q \\ q-r & r-p & p-q \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + \mathrm{R3}$:

$$= \begin{vmatrix} 1 & 3 & 6 \\ p+q+r & p+q+r & p+q+r \\ q-r & r-p & p-q \end{vmatrix}$$

Factor $(p+q+r)$ out of R2:

$$= (p+q+r)\begin{vmatrix} 1 & 3 & 6 \\ 1 & 1 & 1 \\ q-r & r-p & p-q \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - 3\mathrm{C1}$:

$$= (p+q+r)\begin{vmatrix} 1 & 0 & 6 \\ 1 & -2 & 1 \\ q-r & 4r-p-3q & p-q \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - 6\mathrm{C1}$:

$$= (p+q+r)\begin{vmatrix} 1 & 0 & 0 \\ 1 & -2 & -5 \\ q-r & 4r-p-3q & p-7q+6r \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \frac{5}{2}\mathrm{C2}$:

$$= (p+q+r)\begin{vmatrix} 1 & 0 & 0 \\ 1 & -2 & 0 \\ q-r & 4r-p-3q & \frac{7}{2}p+\frac{1}{2}q-4r \end{vmatrix}$$

Then:

$$= (p+q+r)\left(1 \times (-2) \times \left(\frac{7}{2}p+\frac{1}{2}q-4r\right)\right)$$

$$= (p+q+r)(8r-7p-q)$$

23) d)

$$\begin{vmatrix} p-2r & q-2p & r-2q \\ 2 & 4 & 8 \\ q+3r & r+3p & p+3q \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} + \mathrm{R3}$:

$$= \begin{vmatrix} p+q+r & p+q+r & p+q+r \\ 2 & 4 & 8 \\ q+3r & r+3p & p+3q \end{vmatrix}$$

Factor $(p+q+r)$ out of R1:

$$= (p+q+r)\begin{vmatrix} 1 & 1 & 1 \\ 2 & 4 & 8 \\ q+3r & r+3p & p+3q \end{vmatrix}$$

Let the new $\mathrm{C2} = \mathrm{C2} - 2\mathrm{C1}$:

$$= (p+q+r)\begin{vmatrix} 1 & -1 & 1 \\ 2 & 0 & 8 \\ q+3r & 3p-2q-5r & p+3q \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - 4\mathrm{C1}$:

$$= (p+q+r)\begin{vmatrix} 1 & -1 & -3 \\ 2 & 0 & 0 \\ q+3r & 3p-2q-5r & p-q-12r \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - 3\mathrm{C2}$:

$$= (p+q+r)\begin{vmatrix} 1 & -1 & 0 \\ 2 & 0 & 0 \\ q+3r & 3p-2q-5r & 5q+3r-8p \end{vmatrix}$$

Then:

$$= (p+q+r)\big((-2)\times(-1)\times(5q+3r-8p)\big)$$

$$= 2(p+q+r)(5q+3r-8p)$$

24)

$$xy\begin{vmatrix}1 & 2 & 5\\0 & 1 & -3\\0 & 0 & 1\end{vmatrix}$$

Multiply C2 by x:

$$= y\begin{vmatrix}1 & 2x & 5\\0 & x & -3\\0 & 0 & 1\end{vmatrix}$$

Multiply C3 by y:

$$= \begin{vmatrix}1 & 2x & 5y\\0 & x & -3y\\0 & 0 & y\end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} + \mathrm{C3}$:

$$= \begin{vmatrix}1+5y & 2x & 5y\\-3y & x & -3y\\y & 0 & y\end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R2}$:

$$= \begin{vmatrix}1+5y & 2x & 5y\\-3y & x & -3y\\4y & -x & 4y\end{vmatrix}$$ as required

25)

$$xy(x-1)(y-1)\begin{vmatrix}1 & 0 & 1\\ -1 & 1 & -1\\ 0 & -1 & 1\end{vmatrix}$$

$$=\begin{vmatrix}x(y-1) & 0 & y\\ -x(y-1) & x-1 & -y\\ 0 & -(x-1) & y\end{vmatrix}$$

$$=\begin{vmatrix}xy-x & 0 & y\\ x-xy & x-1 & -y\\ 0 & 1-x & y\end{vmatrix}$$

Let the new $\mathrm{C2}=\mathrm{C2}+\mathrm{C3}$:

$$=\begin{vmatrix}xy-x & y & y\\ x-xy & x-y-1 & -y\\ 0 & 1-x+y & y\end{vmatrix}$$

Let the new $\mathrm{R3}=\mathrm{R3}+\mathrm{R2}$:

$$=\begin{vmatrix}xy-x & y & y\\ x-xy & x-y-1 & -y\\ x-xy & 0 & 0\end{vmatrix}$$

Let the new $\mathrm{C3}=\mathrm{C3}+\mathrm{C1}$:

$$=\begin{vmatrix}xy-x & y & xy-x+y\\ x-xy & x-y-1 & x-xy-y\\ x-xy & 0 & x-xy\end{vmatrix}$$ as required

26)

$\mathbf{A}$ is singular $\Rightarrow |\mathbf{A}| = 0$

$$\begin{vmatrix} 6 & 3x+3a+5 & -5 \\ 5 & 4x+4a+4 & -4 \\ 4a-x+1 & 3a-2x+1 & x-4a-1 \end{vmatrix} = 0$$

Let the new $\mathrm{C1} = \mathrm{C1} + \mathrm{C3}$:

$$\Rightarrow \begin{vmatrix} 1 & 3x+3a+5 & -5 \\ 1 & 4x+4a+4 & -4 \\ 0 & 3a-2x+1 & x-4a-1 \end{vmatrix} = 0$$

Let the new $\mathrm{R2} = \mathrm{R2} - \mathrm{R1}$:

$$\Rightarrow \begin{vmatrix} 1 & 3x+3a+5 & -5 \\ 0 & x+a-1 & 1 \\ 0 & 3a-2x+1 & x-4a-1 \end{vmatrix} = 0$$

Let the new $\mathrm{C2} = \mathrm{C2} + \mathrm{C3}$:

$$\Rightarrow \begin{vmatrix} 1 & 3x+3a & -5 \\ 0 & x+a & 1 \\ 0 & -x-a & x-4a-1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 1 & 3(x+a) & -5 \\ 0 & x+a & 1 \\ 0 & -(x+a) & x-4a-1 \end{vmatrix} = 0$$

Factor $(x+a)$ out of C2:

$$\Rightarrow (x+a)\begin{vmatrix} 1 & 3 & -5 \\ 0 & 1 & 1 \\ 0 & -1 & x-4a-1 \end{vmatrix} = 0$$

Let the new $\mathrm{C3} = \mathrm{C3} - \mathrm{C2}$:

$$\Rightarrow (x+a)\begin{vmatrix} 1 & 3 & -8 \\ 0 & 1 & 0 \\ 0 & -1 & x-4a \end{vmatrix} = 0$$

Then:

$\Rightarrow (x+a)(x-4a) = 0$

$\therefore x = -a$ or $x = 4a$

27)

$\mathbf{A}$ is singular $\Rightarrow |\mathbf{A}| = 0$

$$\begin{vmatrix} -2a & 6x+24a & -x-a \\ 2x-4a & 7x+28a & 3x-5a \\ 3x-7a & 13x+52a & 4x-8a \end{vmatrix} = 0$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C3}$:

$$\Rightarrow \begin{vmatrix} x-a & 6x+24a & -x-a \\ -x+a & 7x+28a & 3x-5a \\ -x+a & 13x+52a & 4x-8a \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-a & 6x+24a & -x-a \\ -(x-a) & 7x+28a & 3x-5a \\ -(x-a) & 13x+52a & 4x-8a \end{vmatrix} = 0$$

Factor $(x-a)$ out of $\mathrm{C1}$:

$$\Rightarrow (x-a)\begin{vmatrix} 1 & 6x+24a & -x-a \\ -1 & 7x+28a & 3x-5a \\ -1 & 13x+52a & 4x-8a \end{vmatrix} = 0$$

Let the new $\mathrm{R2} = \mathrm{R2} + \mathrm{R1}$:

$$\Rightarrow (x-a)\begin{vmatrix} 1 & 6x+24a & -x-a \\ 0 & 13x+52a & 2x-6a \\ -1 & 13x+52a & 4x-8a \end{vmatrix} = 0$$

Let the new $\mathrm{R3} = \mathrm{R3} + \mathrm{R1}$:

$$\Rightarrow (x-a)\begin{vmatrix} 1 & 6x+24a & -x-a \\ 0 & 13x+52a & 2x-6a \\ 0 & 19x+76a & 3x-9a \end{vmatrix} = 0$$

Let the new $\mathrm{R3} = \mathrm{R3} - \mathrm{R2}$:

$$\Rightarrow (x-a)\begin{vmatrix} 1 & 6x+24a & -x-a \\ 0 & 13x+52a & 2x-6a \\ 0 & 6x+24a & x-3a \end{vmatrix} = 0$$

Let the new $\mathrm{R2} = \mathrm{R2} - 2\mathrm{R3}$:

$$\Rightarrow (x-a)\begin{vmatrix} 1 & 6x+24a & -x-a \\ 0 & x+4a & 0 \\ 0 & 6x+24a & x-3a \end{vmatrix} = 0$$

Then:

$\Rightarrow (x-a)(x+4a)(x-3a) = 0$

$\therefore x = a$ or $x = -4a$ or $x = 3a$

28) a)

We can let the new $\mathrm{C3} = \mathrm{C3} - p\mathrm{C1}$ and so:

$$\begin{vmatrix} 0 & 1 & x-p \\ x & p & px \\ p & q & p^2 \end{vmatrix} = \begin{vmatrix} 0 & 1 & x-p \\ x & p & 0 \\ p & q & 0 \end{vmatrix}$$

28) b)

$\mathbf{A}$ is singular $\Rightarrow |\mathbf{A}| = 0$

$$\begin{vmatrix} 0 & 1 & x-p \\ x & p & 0 \\ p & q & 0 \end{vmatrix} = 0$$

$\Rightarrow (x-p)(qx-p^2) = 0$

$\Rightarrow x = p$ or $x = \frac{p^2}{q}$

29)

$$\begin{vmatrix} x^3 & 0 & p^2x \\ q & pq & q \\ x+q & p^2+pq & p+q \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} - \mathrm{C3}$:

$$= \begin{vmatrix} x^3 - p^2x & 0 & p^2x \\ 0 & pq & q \\ x-p & p^2+pq & p+q \end{vmatrix}$$

$$= \begin{vmatrix} x(x^2-p^2) & 0 & p^2x \\ 0 & pq & q \\ x-p & p^2+pq & p+q \end{vmatrix}$$

$$= \begin{vmatrix} x(x-p)(x+p) & 0 & p^2x \\ 0 & pq & q \\ x-p & p^2+pq & p+q \end{vmatrix}$$

Factor $(x-p)$ out of $\mathrm{C1}$:

$$= (x-p)\begin{vmatrix} x(x+p) & 0 & p^2x \\ 0 & pq & q \\ 1 & p^2+pq & p+q \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - \frac{1}{p}\mathrm{C2}$:

$$= (x-p)\begin{vmatrix} x(x+p) & 0 & p^2x \\ 0 & pq & 0 \\ 1 & p^2+pq & 0 \end{vmatrix}$$

Factor p^2x out of $\mathrm{C3}$:

$$= p^2x(x-p)\begin{vmatrix} x(x+p) & 0 & 1 \\ 0 & pq & 0 \\ 1 & p^2+pq & 0 \end{vmatrix}$$

Then:

$$= p^2x(x-p)(0 \times (p^2+pq) - pq \times 1)$$

$$= -p^3qx(x-p)$$

$$= p^3qx(p-x)$$

30)

$$\begin{vmatrix} q-x & -p^2 & x-q \\ x-q & px & q-p \\ x-q & p^2 & p+2q-x \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \mathrm{R1}$:

$$= \begin{vmatrix} q-x & -p^2 & x-q \\ x-q & px & q-p \\ 0 & 0 & p+q \end{vmatrix}$$

$$= \begin{vmatrix} -(x-q) & -p^2 & x-q \\ x-q & px & q-p \\ 0 & 0 & p+q \end{vmatrix}$$

Factor $(x-q)$ out of $\mathrm{C1}$:

$$= (x-q)\begin{vmatrix} -1 & -p^2 & x-q \\ 1 & px & q-p \\ 0 & 0 & p+q \end{vmatrix}$$

Let the new $\mathrm{R2} = \mathrm{R2} + \mathrm{R1}$:

$$= (x-q)\begin{vmatrix} -1 & -p^2 & x-q \\ 0 & px-p^2 & q-p \\ 0 & 0 & p+q \end{vmatrix}$$

$$= (x-q)\begin{vmatrix} -1 & -p^2 & x-q \\ 0 & p(x-p) & q-p \\ 0 & 0 & p+q \end{vmatrix}$$

Then:

$$= (x-q)(p+q) \times -p(x-p)$$

$$= p(p+q)(p-x)(x-q)$$

31) a)

$$\begin{vmatrix} a-2x+b & x+a & 4a-2x+b \\ a-2x+2b-1 & x+a+1 & 4a-2x+3b \\ 3a+b-x-1 & 1 & 6a+2b-2x \end{vmatrix}$$

Let the new $\mathrm{R1} = \mathrm{R1} - \mathrm{R2}$:

$$= \begin{vmatrix} 1-b & -1 & -2b \\ a-2x+2b-1 & x+a+1 & 4a-2x+3b \\ 3a+b-x-1 & 1 & 6a+2b-2x \end{vmatrix}$$

Let the new $\mathrm{C1} = \mathrm{C1} + \mathrm{C2}$:

$$= \begin{vmatrix} -b & -1 & -2b \\ 2a-x+2b & x+a+1 & 4a-2x+3b \\ 3a+b-x & 1 & 6a+2b-2x \end{vmatrix}$$

Let the new $\mathrm{R3} = \mathrm{R3} + \mathrm{R1}$:

$$= \begin{vmatrix} -b & -1 & -2b \\ 2a-x+2b & x+a+1 & 4a-2x+3b \\ 3a-x & 0 & 6a-2x \end{vmatrix}$$

$$= \begin{vmatrix} -b & -1 & -2b \\ 2a-x+2b & x+a+1 & 4a-2x+3b \\ 3a-x & 0 & 2(3a-x) \end{vmatrix}$$

Factor $(3a-x)$ out of R3:

$$= (3a-x)\begin{vmatrix} -b & -1 & -2b \\ 2a-x+2b & x+a+1 & 4a-2x+3b \\ 1 & 0 & 2 \end{vmatrix}$$

Let the new $\mathrm{C3} = \mathrm{C3} - 2\mathrm{C1}$:

$$= (3a-x)\begin{vmatrix} -b & -1 & 0 \\ 2a-x+2b & x+a+1 & -b \\ 1 & 0 & 0 \end{vmatrix}$$

Factor $-b$ out of C3:

$$= -b(3a-x)\begin{vmatrix} -b & -1 & 0 \\ 2a-x+2b & x+a+1 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$

Rotate the rows one place downwards:

$$= -b(3a-x)\begin{vmatrix} 1 & 0 & 0 \\ -b & -1 & 0 \\ 2a-x+2b & x+a+1 & 1 \end{vmatrix}$$

Then:

$= -b(3a-x) \times (-1)$

$= b(3a-x)$

31) b)

$a = 6$

$6 + b = 13 \Rightarrow b = 7$

$|\mathbf{A}| = \pm\frac{1260}{300} = \pm\frac{21}{5}$

If $|\mathbf{A}| = \frac{21}{5}$:

$\Rightarrow 7(3 \times 6 - x) = \frac{21}{5}$

$\Rightarrow 18 - x = \frac{3}{5}$

$\Rightarrow x = \frac{87}{5}$

If $|\mathbf{A}| = -\frac{21}{5}$:

$\Rightarrow 7(3 \times 6 - x) = -\frac{21}{5}$

$\Rightarrow 18 - x = -\frac{3}{5}$

$\Rightarrow x = \frac{93}{5}$

$\therefore x = \frac{87}{5}$ or $x = \frac{93}{5}$